JCP campaign notebook

Continuous coordinate-network bases for Petrov–Galerkin reduced models

Tunable-NM-ROM project

Rebuilt October 15, 2026

What this is. The working notebook for the JCP paper: one entry per update, newest last, built by `notebook/build.sh`. It records decisions, designs, results (generated, never typed), and anything retracted. The canonical chronology stays in `LAB-LOG.md`; the plan is `reports/2026-10-06-jcp-offmesh-paper-plan.md`; the outline is `reports/2026-10-06-jcp-combined-paper-outline/`.

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1 2026-10-06 — Campaign start and Phase 0

Thesis

A reduced trial basis that is a *continuous, importance-ordered* function of position, $u(x) = G(x)c$, gives two properties. **Ordered:** one trained model offers an accuracy/cost dial at deployment (width R', head on or off). **Continuous:** the nonlinear term is an integral, so it can be hyper-reduced by classical quadrature (tensor Gauss, rank-1 lattices) at points that ignore the mesh, instead of quadrature fitted to snapshots on the mesh. The paper combines the discarded ICLR material (the dial) with Hari's off-mesh quadrature study. The ICLR paper will not be submitted.

Decisions (user, 2026-10-06)

topic	decision
Paper	One combined JCP paper; ICLR dropped.
Off-mesh code	Copied byte-identically into each lane's `vendor/` with `PROVENANCE.json` (not forked from the quadrature branches).
Full-order comparators	Run *and compare against* all three of coarse-grid, second-order and spectral solvers, in addition to Newton–BiCGStab/GMRES and IMEX-CG. Supersedes the 2026-09-21 "named iterative only" rule.
Examples	Keep the current PDEs and add harder ones (chosen in the lane designs).
NS 3D	Keep the lane and keep improving its coordinate-network bank.
Worktrees	Claude names and creates them without asking (now in `CLAUDE.md` and `AGENTS.md`).
Notebook	This LaTeX notebook, rebuilt after every update.

Phase 0 checklist

- [done] Plan and five planning notes, revised after a Codex audit: `reports/2026-10-06-jcp-offmesh-paper-plan.md`.
- [done] Paper outline (PDF): `reports/2026-10-06-jcp-combined-paper-outline/`.
- [done] Literature check of the unread preprints and a 2024–2026 sweep (plan appendix 07); see below.
- [done] Week-1 lanes created (table below).
- [done] `CLAUDE.md`, `AGENTS.md` and memory updated with the decisions.
- [user] Withdraw the ICLR submission on OpenReview if it is still active.
- [user] Agree authorship and the split of work with Hari; get his consent to use his code and results.

Literature check: outcome

None of arXiv 2505.14595 (Φ-ROM, a grid-based contrast), 2507.07830 (meshless SPH projection, no hyper-reduction), 2508.21279 (energy-conservative EQP, a fitted method) or the LiCROM cubature (random mesh vertices, equal weights) pre-empts the claims. No ROM paper found uses weighted classical quadrature for the reduced residual. Per contribution:

- ordered tunable bank: **novel for ROMs**; credit nested dropout (Rippel et al. 2014), slimmable networks, POD truncation, LiCROM;
- classical quadrature replacing fitted hyper-reduction: **novel as a combination**; ingredients partly pre-empted (Monte Carlo/uniform points in Simplicits, SNF-ROM, Weder et al.; lattices in PINN training; fitted off-mesh points in CECM);
- decoupling from the mesh's $O(h)$ error: **partly pre-empted** (SNF-ROM has the mechanism); Ingimarson–Rebholz–Iliescu (CMAME 2022) argue the opposite position and must be addressed;
- rule-family behaviour on tensor sine tests: **novel in the ROM setting**, as known theory with a new consequence;
- data-free error indicator: **partly pre-empted in principle**; call it an indicator, never a certificate.

Still to read before submission: arXiv 2607.27471, 2607.10965, 2602.01397, 2504.15657.

Week-1 lanes

All fork from `exp/2026-10-01-ns3d-coordnet-bank` @ `21c175a1b`; each mirrors to `origin/codeonly/exp/<lane>` with its own `sync_github.sh`, and vendors 69 files of off-mesh code, each checked blob by blob against its source commit (`exp/2026-10-01-quadrature-study` @ `84d28e303`, `exp/2026-10-01-quadrature-burgers3d` @ `aab6b6927`).

lane (`worktrees/`, branch `exp/...`)	experiments	role
`2026-10-06-jcp-references`	E0	order-verified second-order references; pseudo-spectral cross-check; sealed and out-of-distribution cohorts in a read-only store
`2026-10-06-jcp-mechanism`	E1, E2	rule ladders against theory; second-order stencil tensor, mesh-node analytic gradient, coarse-trained bank, all-continuum arm
`2026-10-06-jcp-fom-pareto`	E4	full-order cost ladders (iterative, IMEX-CG, second-order, coarse-grid, spectral); timing harness
`2026-10-06-jcp-hr-baselines`	E5	POD-LSPG/Galerkin with ECSW, Q-DEIM, GNAT; POD + exact tensor; best-practice EQ; CROM-style sampler

Harder examples: candidates for the designs

- time-dependent reaction–diffusion with e^u (Frank-Kamenetskii): non-polynomial, so no tensor exists;
- porous-medium diffusion $\nabla \cdot (u^m \nabla u)$ with non-integer m;
- Burgers at lower viscosity / narrower bumps, to find where the rule size and bank floor break;
- a non-box domain (e.g. L-shape with a distance-function Dirichlet factor and a composite rule);
- NS 3D with an improved coordinate-network bank.

Carried-forward corrections (from the Codex audit of the plan)

Retracted: the explicit Gauss strip-width bound in planning note 03; "lattices converge like m^{-3}" as an asymptotic rate (only a conditional upper bound); Hari's "super-algebraic lattice" and "tent adds a kink" statements for this integrand; the per-lane sealing of test cohorts (now global); the 5 % timing fault under a 10 % gate; Gauss "exactness" on sines; the rule-doubling "certificate"; the 5–7-day schedule (at least 7.6 days on four GPUs).

Next (Phase 1)

For each week-1 lane: write `DESIGN.md` → Codex design audit → code → Codex code audit → real-size smoke with every control firing → cluster jobs → Codex results audit → generated report and a notebook entry. Lanes run as separate subagents, one worktree each, with distinct cluster namespaces.

2 2026-10-08 — Launch: accuracy lanes (C) and two mechanism checks (A)

Decision (user)

Push accuracy first: run the three Group C directions, and from Group A only the two most important checks. Every step in every lane is gated by a Codex adversarial check (user instruction); the lane rules are in `reports/2026-10-08-jcp-lane-agent-rules.md`.

Lanes running

lane	question	cluster cap
`2026-10-08-jcp-wide-bank` (C1)	Now that off-mesh memory is linear in R', how far does a wider bank push accuracy? 2D at $R' = 512$; trimming the test count M; training a 3D bank with $R = 1024$; the error/cost scaling law.	1 job
`2026-10-08-jcp-time2` (C2)	Does second-order time stepping (Crank–Nicolson, BDF2) remove the backward-Euler error and allow larger steps? Includes a fair second-order full solver.	1 job
`2026-10-08-jcp-smooth-bank` (C3)	Does a bank trained for its derivatives (Sobolev loss, smoother features) lower the error floor and need fewer quadrature points? Includes a coarse-data control.	1 job
`2026-10-06-jcp-mechanism` (A1, A2)	Is the gain from leaving the mesh or from the exact gradient (mesh nodes with the analytic gradient)? Does the old error follow the stencil order (upwind vs central gap slope)? A3/A4 deferred.	1 job

Corrections recorded

Retracted: "Our trained bank already has more columns than we used" was wrong for 3D: the 3D bank has exactly 512 columns, all used. A wider 3D bank must be trained; only 2D has headroom (384 of 512 used).

Status

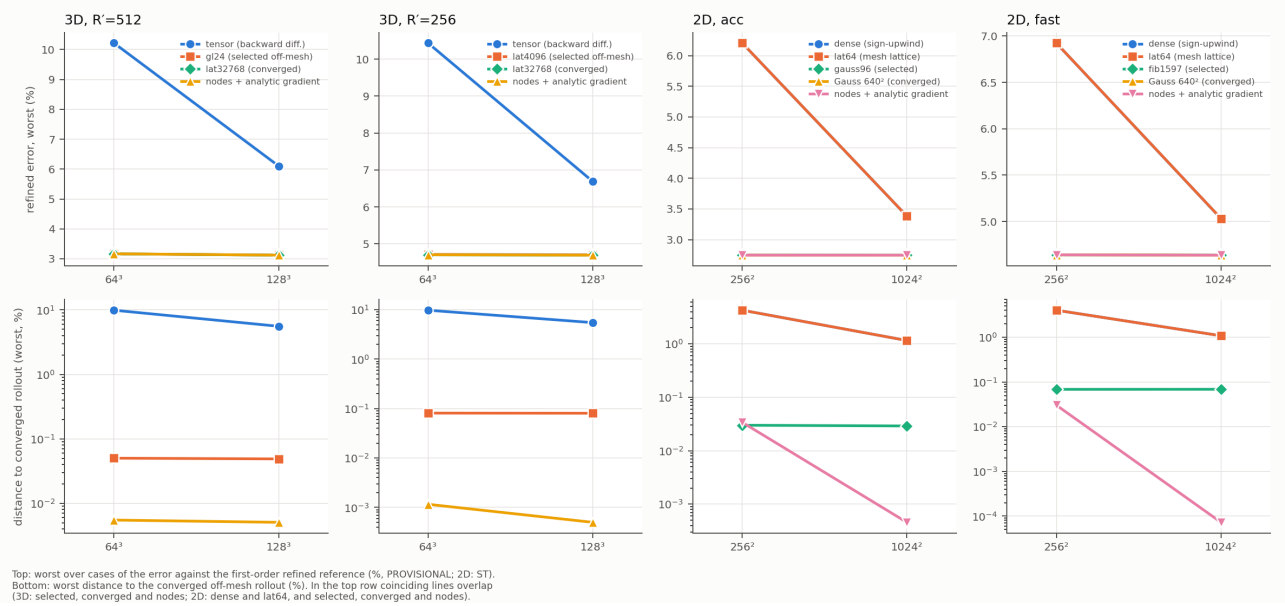
All accuracy numbers from these lanes are *provisional* until the second-order references (E0) exist. The Group B lanes (references, full-solver cost ladder, POD baselines) are not started.

3 2026-10-08 — Mechanism: the gain is the exact gradient (A1, A2)

Lane `2026-10-06-jcp-mechanism` (last commit `712d682d8`, mirror `977b28357`); jobs 5012942 (A1 3D), 5015576 (A1 2D), 5016189 (A2), all on H200, f64, `jax_backend=gpu`. Every step passed a Codex gate; the final report audit had no WRONG items. Report: `worktrees/2026-10-06-jcp-mechanism/experiments/jcp-mechanism/report.md`. **Preliminary; accuracy against first-order references is provisional.**

A1: mesh nodes with the exact gradient

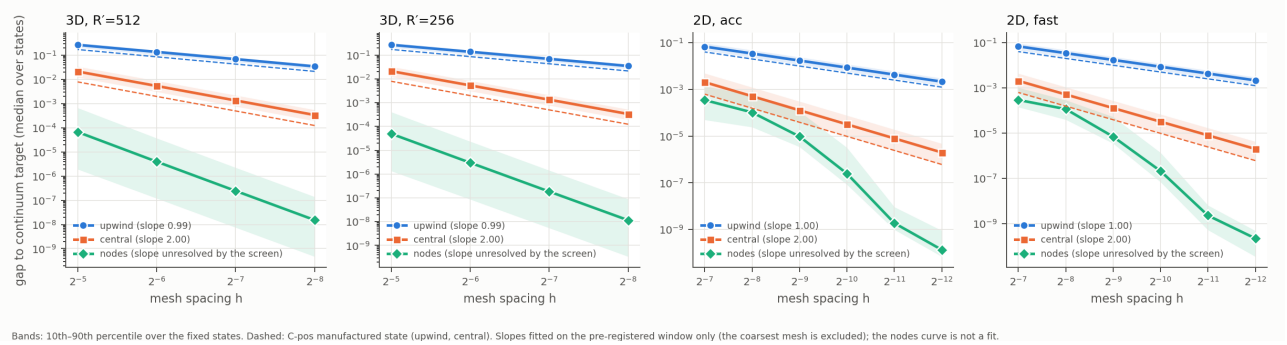
A control arm keeps every mesh node but uses the bank's analytic derivative instead of the upwind stencil. In 3D (64^3, 128^3; $R' = 512,256$; 64 validation cases) it reproduces the converged off-mesh rollout in all four cells (pre-registered label R): the old tensor's worst error is about 10 % (64^3) and 6 % (128^3), while the mesh-node arm, Gauss 24^3 and the 32768-point lattice all sit near 3.1 %. In 2D the pre-registered rule could not be applied (label X0: the dense-vs-converged gap was below the 1 % threshold), but descriptively the mesh-node arm is again far closer to the converged rollout than the dense stencil solve.



Top: worst over cases of the error against the first-order refined reference (%, PROVISIONAL; 2D: ST).
Bottom: worst distance to the converged off-mesh rollout (%). In the top row coinciding lines overlap (3D: selected, converged and nodes; 2D: dense and lat64, and selected, converged and nodes).

A2: the stencil gap follows the stencil order

On fixed saved states, the gap between the mesh sum and the continuum integral falls with slope 0.99–1.00 for the first-order upwind stencil and 2.00 for second-order central differences, inside the pre-registered bars in all four panels, with both controls behaving. With the exact gradient the gap reaches the reference's noise floor, so its slope is not resolved.



Bands: 10th–90th percentile over the fixed states. Dashed: C-pos manufactured state (upwind, central). Slopes fitted on the pre-registered window only (the coarsest mesh is excluded); the nodes curve is not a fit.

What this changes in the paper

Kill criterion K1 fired in its useful form. The accuracy gain comes from evaluating the continuum term with the exact gradient, not from placing points off the mesh.

- The claim becomes: exact-gradient *continuum evaluation* gives the accuracy.
- Classical quadrature (thousands of points, independent of the mesh) gives it cheaply. The mesh-node arm needs every node: about 2.5×10^5 at 64^3 and 2×10^6 at 128^3.

Changed or corrected along the way

- The 2D finite-difference gradient gate was redesigned as a convergence check (the bank is steep near the walls; the analytic gradient is correct).
- Historical cohort hashes are now report-only, because they include GPU floats.
- A non-finite target now invalidates a result rather than being retried.
- A halved A2 state set was caught by the code audit before the job ran.

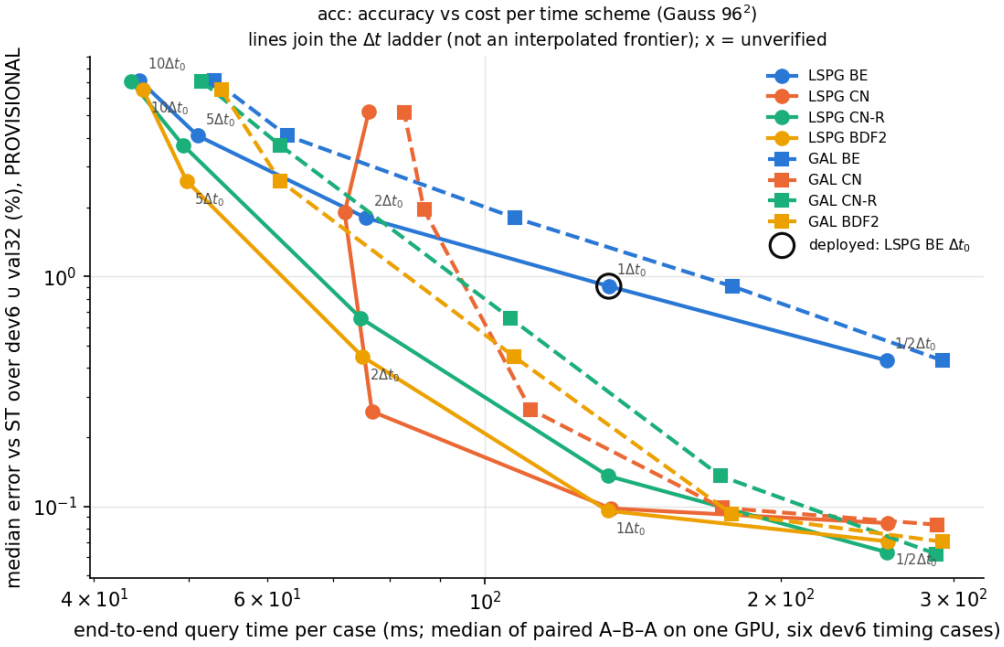
Open: whether to pre-register a new 2D separation rule, and a solver-sensitivity rerun at 128^3. A3 and A4 remain deferred.

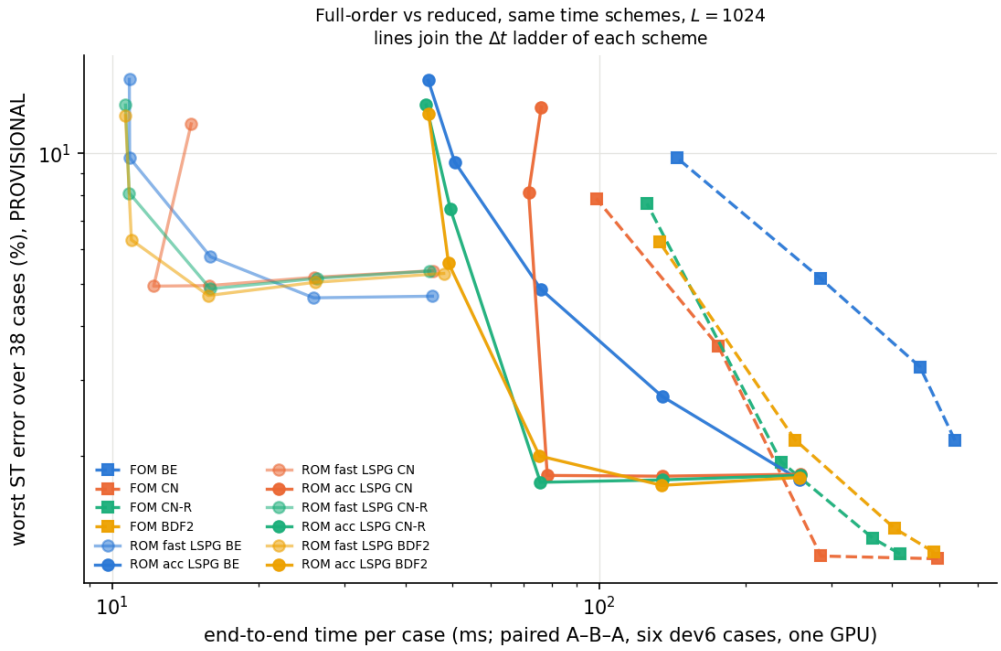
4 2026-10-08 — Second-order time stepping (C2, complete)

Lane `2026-10-08-jcp-time2` (last commit `a7a12528f`, mirror `1445e4735`). Jobs 5012794, 5012974, 5015584, 5017631 (A100), 5020049, 5023437 (H200); all audited, namespace emptied. Every step was Codex-gated, and the final report audit came back clean. Report: `worktrees/2026-10-08-jcp-time2/experiments/jcp-time2/report.md`. **All accuracy numbers are provisional** (first-order references).

2D, accurate setting ($R' = 384$, Gauss 96^2, 1024^2, 38 cases)

- Same step and cost: worst/median error vs ST goes from 2.747/0.906 % (backward Euler) to 1.796/0.098 % (Crank–Nicolson). Time error falls from 0.978 % to 0.026 %.
- At twice the step, Crank–Nicolson gives 1.803/0.259 % and runs about 1.75× faster. Both pre-registered hypotheses pass.
- Fair comparison: the full solver given the same schemes is also faster and more accurate with Crank–Nicolson. Matched speed-up against it is 1.76× (backward Euler) and 3.65× (Crank–Nicolson at $2\Delta t$).





Other settings

- Wide ($R' = 512$): 2.837/0.907 % → 2.166/0.111 %; both hypotheses pass.
- Fast ($R' = 128$): the median improves (1.035 → 0.419 %) but the worst case worsens (4.637 → 5.171 %); both fail, because the small bank's error dominates.
- 3D (65^3, 16 cases): time error drops 21× (1.87 → 0.089 %) at about equal cost. The generic solver is about 1.7× the deployed fixed sweep, so the gain is accuracy, not speed. The 3D reference is itself backward Euler at $\Delta t/4$, so it cannot rank the schemes (pre-registered).

Retracted or corrected

Retracted: the "1–2 percentage points" of backward-Euler time error is the maximum gap; the median gap is 0.5–0.8 pp. Least-squares Crank–Nicolson is not second order and must not be called so; the Galerkin form and the CN-R/BDF2 least-squares variants are. Comparing against a backward-Euler full solver overstates the speed-up.

Open

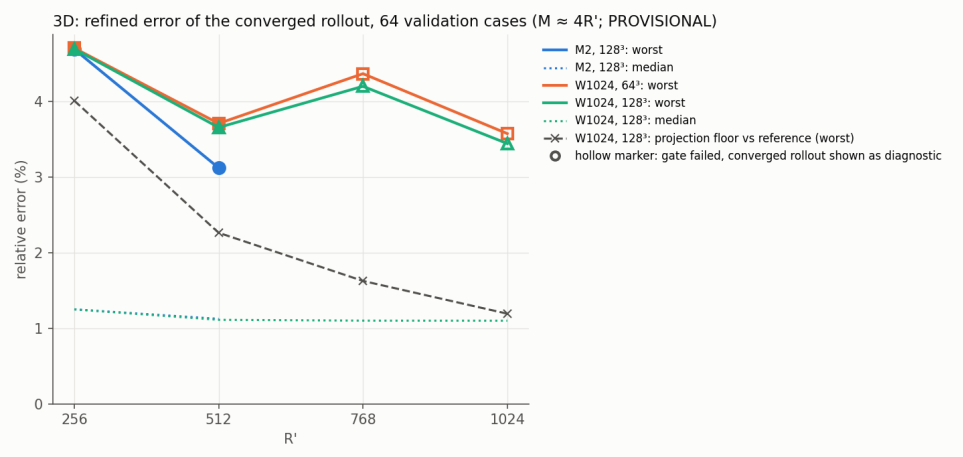
A fixed-sweep second-order solver for 3D (the route to a 3D speed-up); rerun against second-order references; choose the Galerkin or least-squares form for the paper; test64 and 4096^2 replication.

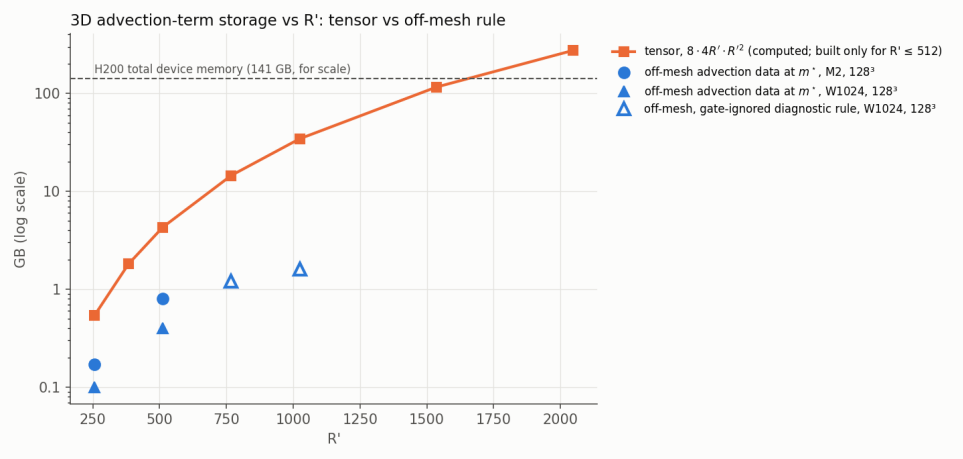
5 2026-10-09 — Wider bank (C1, complete): memory wall gone, accuracy gain not yet

Lane `2026-10-08-jcp-wide-bank` (last commit `2c3c397e0`, mirror `dbb3da96f`). Jobs 5012313, 5012763 (A100); 5012847, 5022387 (H200); 5020724, 5020855, 5021867, 5022113 (H100). Every step was Codex-gated, and the final report audit says REPORT-OK. Report: `worktrees/2026-10-08-jcp-wide-bank/experiments/jcp-wide-bank/report.md`. **Provisional:** first-order references, and the 3D selections are post hoc (A7). The J4 solver uses 6 adaptive steps (A8), so J4 is not comparable with earlier 3D runs.

Findings

- **2D dial stops at $R' = 384$.** Against the space-only reference, the worst error falls 3.45 → 1.49 → 1.05 % for $R' = 128 \to 256 \to 384$, but $R' = 512$ is worse on 28 of 38 cases at 3.4× the cost. Against the space+time reference the median is flat at 0.91 %, consistent with time-step error dominating (a hypothesis).
- **Test-count trim does not work.** The registered mechanism is contradicted in 2D and 3D; only one 2D trim gives a 25 % saving for a negligible ST change.
- **Wider 3D bank ($R = 1024$) trained.** Its worst projection floor falls 3.23 → 1.93 → 1.29 % for $R' = 512 \to 768 \to 1024$.
- **But the 3D ROM barely follows.** From $R' = 512$ to 1024 every case improves slightly, yet the median stays near 1.1 % and the worst error is non-monotone (about 3.4–4.4 %). The registered verdict is *unavailable*, because a quadrature-convergence gate fails at $R' \geq 768$.
- **The memory wall is gone.** At $R' = 1024$ the off-mesh rule needs 1.61 GB, against 34.37 GB for the tensor.





Retracted

Retracted: the 4-case preview of 1.43 % worst error at $R' = 1024$, 128^3 (relayed on 2026-10-08) did not hold on all 64 cases: the full-cohort worst is about 3.4 %. "The 2D dial extends to 512" and the registered test-count-trim mechanism are both retracted.

What it means

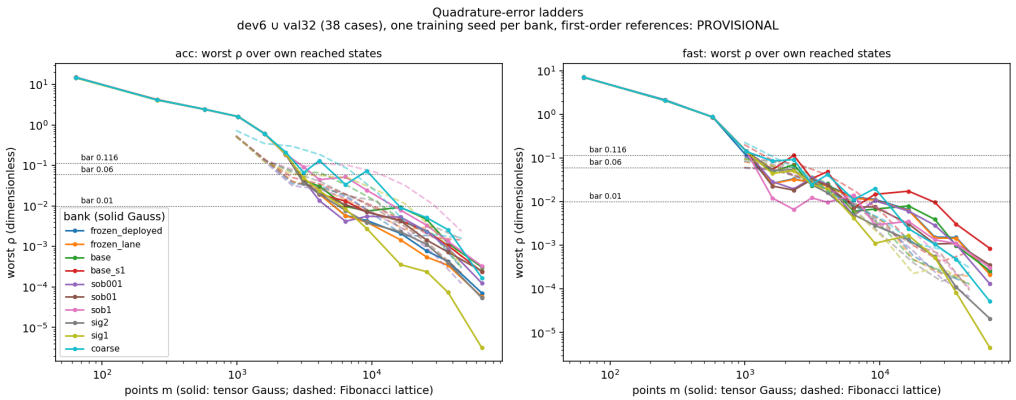
The bank is no longer the bottleneck: the reduced model sits well above the bank's projection floor. With C2 (time error dominates) and C3 (smoother or Sobolev banks do not help the ROM), the evidence points to time discretisation and the first-order reference as the next limits. The obvious next experiment is second-order stepping with the wide bank, against second-order references.

6 2026-10-09 — Derivative-aware bank (C3, complete): the bank is not the bottleneck

Lane `2026-10-08-jcp-smooth-bank`, last commit `12f620782`, mirror `8ede69cba`. Jobs 5016275 and 5027733 (training, 4× A100 each); 5027022 and 5037650 (evaluation). All steps were Codex-gated. The report audit confirmed all fixes except four minor presentation items. Report: `worktrees/2026-10-08-jcp-smooth-bank/experiments/jcp-smooth-bank/report.md`. **Provisional:** first-order references, one training seed per recipe, development and validation cohorts only.

Findings

- **Smoother banks need no fewer points at the acceptance bars.** In the accurate setting every bank needs Gauss 56^2. The point count there is set by the $M = 1536$ high-frequency tests, not by the bank: with only the first 320 tests, Gauss 32^2 is already near the bar.
- **Smoothness pays only in the high-accuracy tail.** At 256^2 points the worst ρ is 3.2×10^{-6} for $\sigma = 1$, against 3.1×10^{-4} for the baseline, with no loss of representation accuracy.
- **Sobolev training improves the bank's derivatives but not the ROM.** The median gradient error falls by 53, 70 and 74 % for $\lambda = 0.01, 0.1, 1$. The ROM error against the space-only reference does not improve (it rises by 62 % at $\lambda = 1$).
- **All banks sit at the time-stepping floor** against the space+time reference: 2.74–2.97 % in the accurate setting.
- **Coarse-data control.** A bank trained only on 129-node data gives 0.83 % worst error against the space-only reference, against 7.58 % for the 129-node FOM and 4.12 % for the 257-node FOM, in the accurate setting. So finer training data were not necessary for the gain. This holds in the accurate setting only, and the cause is not isolated. The same bank needs 16384 points.



Corrections

Retracted: runs/SUMMARY-R3.md describes the wrong training job; the deployed bank is r3a. The 2D lane's training-overlap check used the wrong draw (the cohorts were re-verified as disjoint). Retrains are not bitwise reproducible without deterministic XLA flags. In the fast setting, the 2D numbers depend on the column-ordering procedure (3.45 % vs 4.51 %).

Week summary (8–9 October)

- **Mechanism (A):** the accuracy comes from the exact gradient; classical quadrature makes it cheap.
- **Second-order time stepping (C2):** the clear win. The median error falls about 9× at equal cost, and the matched speed-up against a fair second-order FOM rises from 1.8× to 3.7×.
- **Wider bank (C1)** and **smoother or Sobolev bank (C3):** each represents the solution better, but the ROM error does not follow. The bank is no longer the bottleneck.
- **Next:** second-order stepping with the wide bank, measured against second-order references (the E0 lane); a fixed-sweep second-order 3D solver; a test space with lower mixed frequencies.

7 2026-10-14 — Second-order references (E0, complete): 3D pass, 2D pass with flags

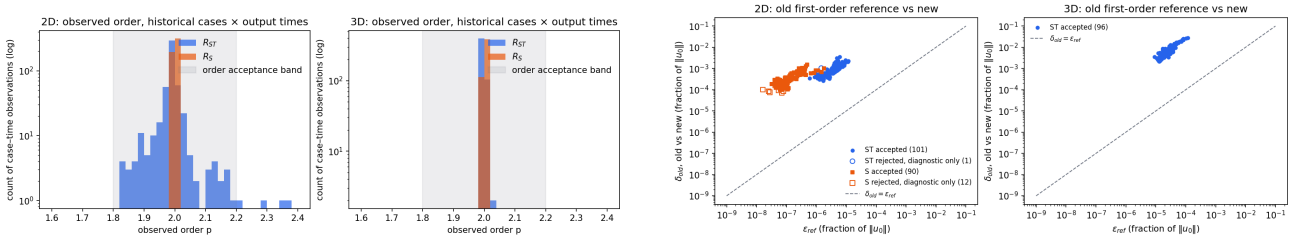
Lane `2026-10-06-jcp-references`, last commit `cc04f4b54`, mirror `66ee55015`. Cluster namespace `jcp-ref`; 39.7 GPU-h (A100 and H200). Every step was Codex-gated, and the final report audit is READY. Report: `worktrees/2026-10-06-jcp-references/experiments/jcp-references/report.md`. Loader: `refstore.py`. Numbers below are quoted from the lane's generated report.

Findings

- **3D passes.** All 200 historical case-kinds pass the order check, with observed order 1.99–2.02. The median ϵ_{ref} is $\approx 2.5 \times 10^{-5}$ and the worst is 1.16×10^{-4}. The pseudo-spectral cross-check passes 8/8, with the measured distance equal to 1.00–1.02 of ϵ_{ref}. Sealed cohort 128/128, OOD 32/32.
- **2D passes with flags.** 191/204 historical case-kinds pass, with $\epsilon_{ref} \approx 3$–4×10^{-6} (space+time) and $\approx 1.2 \times 10^{-7}$ (space only). The 13 flagged cases are not rankable:
  - 12 high-viscosity space-only cases fail the (conservative) solver-floor check;
  - test64 case 36 fails the order check.

  The cross-check passes 8/8. Sealed cohort 230/256 accepted, OOD 64/64.
- **How wrong the old first-order references were** (maximum over output times):
  - 3D: median 0.7 %, maximum 2.81 %;
  - 2D: median 0.03–0.08 %, maximum 0.35 %.

  So 3D numbers can move noticeably on re-scoring, and 2D numbers barely.
- **Controls fired.** A deliberately first-order solver and an injected first-order defect are both rejected. The cross-check against the old references fails on 16/16 cases.



Caveats and corrections

Retracted: Amendment A10 replaced the pre-registered cross-check rule (pseudo-spectral order ≥ 2.5) after the pseudo-spectral data had been seen, though before any comparison with the new references. Under the original rule, only 1/8 2D cross-checks would pass, and 2D would be E0 FAIL. Both verdicts are reported per case. The earlier "1.7–2.3 %" for the old 3D reference error was too low: the maximum is 2.81 %. ϵ_{ref} is a conditional Richardson estimate, not a bound. Sealing is cooperative only: the key sits on the shared cluster account. The sealed cohorts are generated but **not opened**, and no ROM has been scored on them.

Open

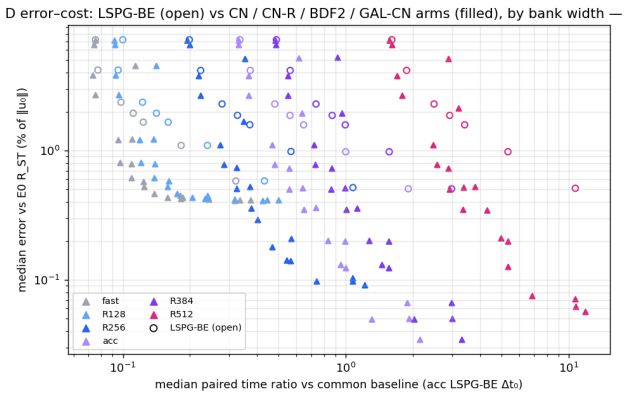
- Key custody.
- Whether to revisit the 2D floor check before opening the sealed cohort.
- Building the opening tool.
- A CN-matched space-only reference.
- Re-scoring every earlier provisional number against this store.

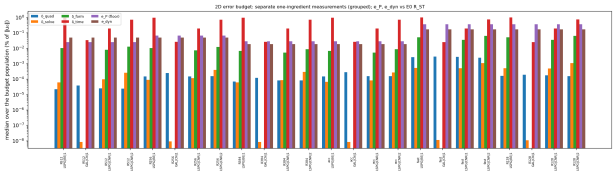
8 2026-10-15 — Step 2: second-order stepping with the wide bank (complete)

Lane `2026-10-13-jcp-combined`, last commit `a3c9e170d`, mirror `c51cecdbd`. Namespace `jcp-comb`, now empty; the jobs are in `runs/JOB-LEDGER.json`. There are 44 Codex audits, and the final report audit is READY. Report: `worktrees/2026-10-13-jcp-combined/experiments/jcp-combined/report.md`. Primary scoring is against the E0 second-order references, with E0's conditional label. First-order scores are kept beside them and stay provisional. The 3D rule selection is **post hoc** (A10, A11). Numbers are quoted from the generated report.

2D (38 development and validation cases)

- **Selected scheme:** LSPG-CN-R. At equal cost, the accurate arm's median error falls from 0.98 % (BE) to 0.20 % (`acc` setting), and the worst from 3.0 % to 1.8 %.
- **Bank width (H1):** R128→R256 passes (median −53.7 %, 38/38 cases improve). R256→R384 fails the 25 % bar (−5.2 %). R384→R512 does not pay off.
- **Tunability (H2):** passes.
- **Speed against the second-order FOM (H5):** the selected scheme's best matched speed-up is 13.6×. The overall best is 17.7×, from an LSPG-CN arm that is not second order. The re-measured C2 `acc` arms reach 3.89×. The FOM is second order in time only.
- **H3:** the error-cancellation explanation for the fast setting's worst case is *not supported*.
- **Finalist quadrature check:** the selected arms fail it at `fast`, `acc` and R256. The fallbacks for the fast arms qualify; those for the accurate arms cost 1.3–1.6× and do not.

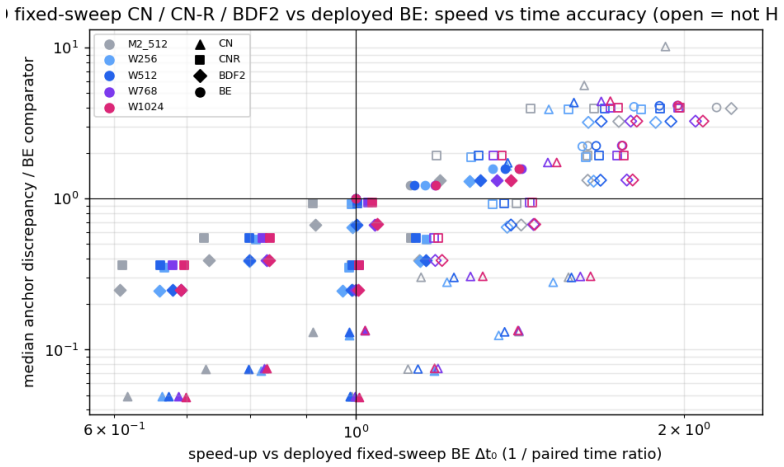




3D (validation cohort 923801, 64 cases, 65^3)

- **Accuracy at equal cost (H4b):** passes at every width. At W1024 the selected arm's median error is 0.12 %, against 1.53 % for the deployed BE. The worst errors of the selected arms are still 4.4–6.7 %.
- **Speed at equal accuracy (H4a, ratio ≤ 0.6):** fails at every width. The closest equal-accuracy arms reach only 0.85–0.96.
- **Width dial (H6):** W512→W768→W1024 extends (median −60 %, then −53 %). W256→W512 does not.

- **J4 gate failure at $R' \geq 768$:** caused by one case (case 5). W1024 converges only at Gauss 80^3.
- **Fixed-sweep CN (K=2, $2.5\Delta t_0$)** fails on a fixed set of 7 cases, so the pre-registered rule selection was unavailable.



What limits next

Time discretisation is still the largest budget component (2D: 16 of 24 setting-arms; 3D: 13 of 19), followed by the bank's projection floor.

Corrections

Retracted: The pre-registered 3D rule selection was unavailable; the one used is post hoc (A10, A11). A first frozen 3D fast-equal-accuracy selection was retracted before any budget job (A15). A first-order H1 and H6 verdicts change under E0 (A13). Earlier drafts used second-order wording for the reduced arms and stacked non-additive budget bars. Offline scoring on the local GB10 GPU corrupted R'=1024 decodes, so all offline analysis now runs on CPU.

Follow-up

The 3D speed lane (`2026-10-14-jcp-speed3d`) is running. It tries one-pass Jacobian reuse at large steps, narrower banks, overhead cuts and batching, and large-mesh FOM comparisons, including a pseudo-spectral FOM.